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B.Tech. / M.Tech. (Integrated) DEGREE EXAMINATION, DECEMBER 2024

First Semester

21BTB103T - BIOLOGY

(For the candidates admitted from the academic year 2024 - 2025)

Note:

- (i) **Part - A** should be answered in OMR sheet within first 40 minutes and OMR sheet should be handed over to hall invigilator at the end of 40th minute.
(ii) **Part - B & Part - C** should be answered in answer booklet.

Time: 3 hours

Max. Marks: 75

PART - A (20 x 1 = 20 Marks)

Answer ALL Questions

Marks BL CO PO

- | | | | | |
|---|---|---|---|---|
| 1. DNA in prokaryotes are present in | 1 | 2 | 1 | 1 |
| A) Mitochondria | | | | |
| B) Ribosomes | | | | |
| C) Plasma membrane | | | | |
| D) Nucleoid region | | | | |
| 2. How many chromosomes are present in human cell? | 1 | 2 | 1 | 1 |
| A) 46 pairs | | | | |
| B) 46 | | | | |
| C) 23 | | | | |
| D) 12 pairs | | | | |
| 3. Calcium ion regulation is done by which part of the cell | 1 | 2 | 1 | 1 |
| A) plasma membrane | | | | |
| B) cytoplasm | | | | |
| C) golgi complex | | | | |
| D) endoplasmic reticulum | | | | |
| 4. Covalent bond that is present in protein that joins amino acids is | 1 | 2 | 2 | 1 |
| A) peptide bond | | | | |
| B) disulphide bond | | | | |
| C) hydrogen bond | | | | |
| D) glucosidic bond | | | | |
| 5. Sugar in DNA molecule is | 1 | 2 | 2 | 1 |
| A) Ribose | | | | |
| B) Deoxyribulose | | | | |
| C) Deoxyribose | | | | |
| D) Glucose | | | | |
| 6. The bond that connects the sugar molecules in starch is | 1 | 2 | 2 | 1 |
| A) peptide bond | | | | |
| B) glycosidic bond | | | | |
| C) phospho diester bond | | | | |
| D) hydrogen bond | | | | |
| 7. Fatty acid is stored as ----- for energy requirement | 1 | 2 | 2 | 1 |
| A) fats | | | | |
| B) phospho lipid | | | | |
| C) wax | | | | |
| D) glycerol | | | | |
| 8. Genetic information is carried from nucleus to cytoplasm through | 1 | 2 | 2 | 2 |
| A) DNA | | | | |
| B) RNA | | | | |
| C) protein | | | | |
| D) fibers | | | | |
| 9. Penicilin kills bacteria by | 1 | 2 | 3 | 1 |
| A) inhibiting cell wall synthesis | | | | |
| B) by stoping replication | | | | |
| C) by inhibiting its motility | | | | |
| D) by inhibiting translation | | | | |
| 10. Tetracyclin resistance to bacteria is by | 1 | 2 | 3 | 1 |
| A) enzyme modification | | | | |
| B) altered targets | | | | |
| C) efflux pumps | | | | |
| D) decreased uptake | | | | |

11. Tetanus Toxoid vaccine is a -----	1	2	3	1
A) protein				
B) killed bacteria				
C) DNA of <i>Closteridium tetani</i>				
D) killed Virus				
12. Antibiotics are used to treat	1	2	3	1
A) Parasite infection				
B) Viral Infection				
C) Bacterial Infection				
D) Life style disease				
13. Enzyme used in glucose monitor is _____	1	2	4	1
A) glucose oxidase				
B) sucrase				
C) peroxidase				
D) catalase				
14. Energy requirement for protein based machines is met by	1	2	4	1
A) ATP				
B) proton motive force				
C) ADP				
D) NAD				
15. Kinesin family of protein moves along	1	2	4	1
A) Myosin				
B) Actin				
C) Dynein				
D) Microtubules				
16. Stents used for angioplasty are made of	1	2	5	1
A) Titanium				
B) Aluminium				
C) Composite				
D) Ceramic				
17. Most commonly used biomaterial in dental application is	1	2	5	1
A) Ceramic				
B) Metal				
C) Polymer				
D) Alloy				
18. Which of the following can mimic the teeth	1	2	5	1
A) Hydroxyapatite				
B) Calcium				
C) Magnesium				
D) Iron				
19. Which of the following natural polymer is used as biomaterial	1	2	5	1
A) Starch				
B) Glycogen				
C) Chitosan				
D) Potato				
20. Which of the property is essential for biomaterial if it is used in bone replacement	1	2	5	1
A) load bearing				
B) thermal conductivity				
C) elasticity				
D) opaque nature				

PART - B (5 x 8 = 40 Marks)
Answer ALL Questions

Marks BL CO PO

21 a. Explain the function of endoplasmic reticulum and its interaction with golgi complex in transport of macromolecules	8	3	1	1
(OR)				
b. Discuss the current and potential future applications of stem cells in regenerative medicine	8	3	1	1
22 a. Discuss various classes of lipids and their functions in biological system	8	2	2	1
(OR)				
b. How do plants produce glucose by photosynthesis? Explain with diagrams	8	2	2	1
23 a. Differentiate attenuated vaccines from subunit vaccines on their mechanism of action	8	3	3	1
(OR)				
b. Discuss the mechanism of bacterial resistance against antibiotics.	8	3	3	1

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|---|---|---|---|---|
| 24 a. Discuss the various types of biosensors with its mechanism | 8 | 3 | 4 | 2 |
| (OR) | | | | |
| b. Explain how ATP is generated by ATP synthase with diagrams | 8 | 3 | 4 | 2 |
| 25 a. How cargo is transported in the cells? Explain with diagrams | 8 | 3 | 4 | 2 |
| (OR) | | | | |
| b. Discuss the properties of biomaterials that are used in medical field. | 8 | 3 | 5 | 2 |

PART - C (1 x 15 = 15 Marks)

Marks BL CO PO

Answer **ANY ONE** Question

- | | | | | |
|--|----|---|---|---|
| 26. Explain the use of stem cells in medical application and elaborate the ethical issue associated with it | 15 | 4 | 1 | 2 |
| 27. Bacteria attain resistance due to overuse of antibiotics. What are all the risk associated with this? How this can be addressed? | 15 | 4 | 3 | 2 |
